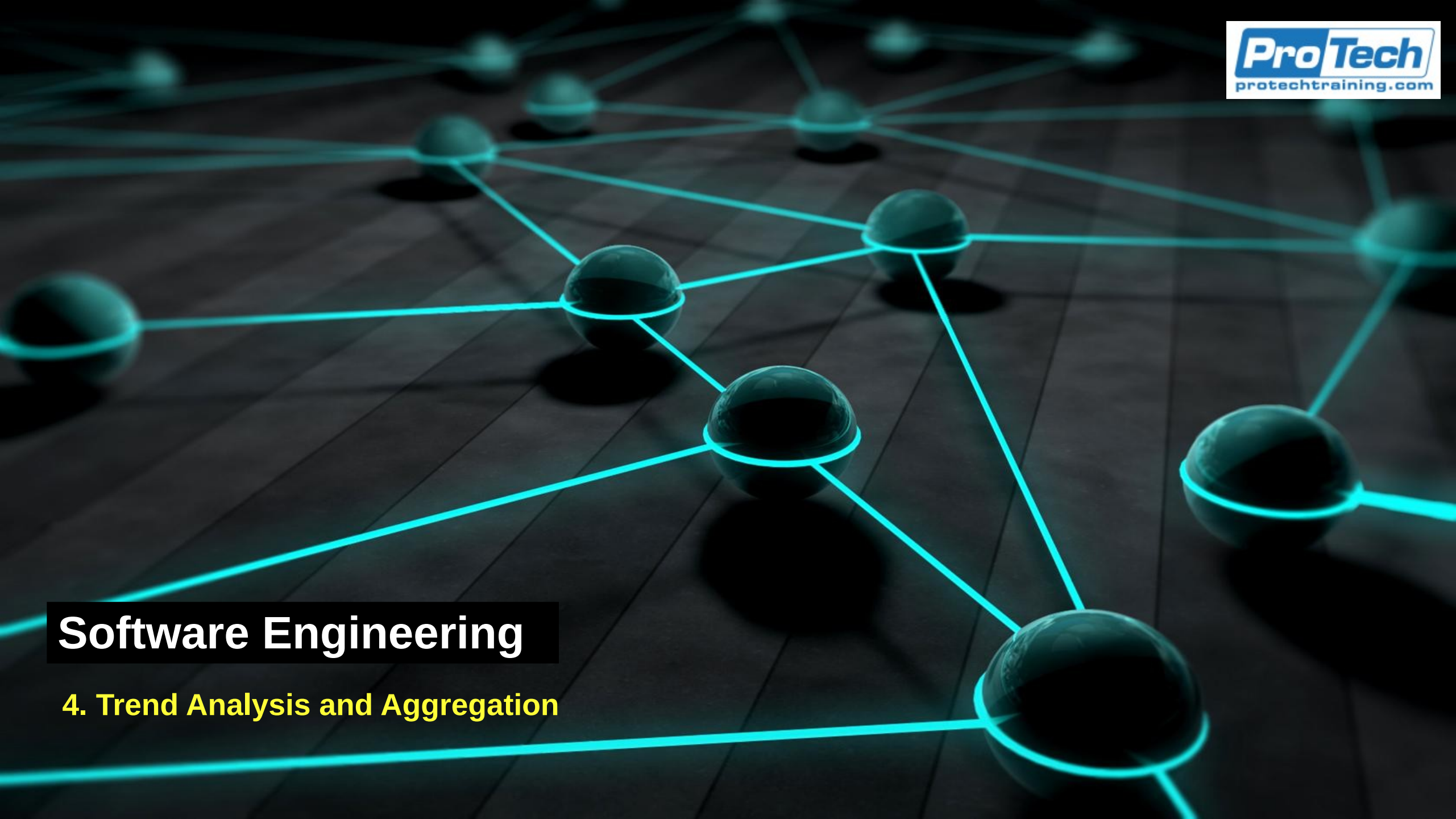




Software Engineering

4. Trend Analysis and Aggregation

KPI and KRI Fundamentals

- In mature risk and operations environments
 - KPIs and KRIs support a resilient organization
 - *Enable monitoring of system health*
 - *Anticipate emerging risks*
 - *Align operational performance with strategic risk appetite*
 - KPIs reflect how well controls and processes are functioning
 - KRIs indicate how monitors thresholds where risk materializes as incidents
 - Together, they enable predictive risk management

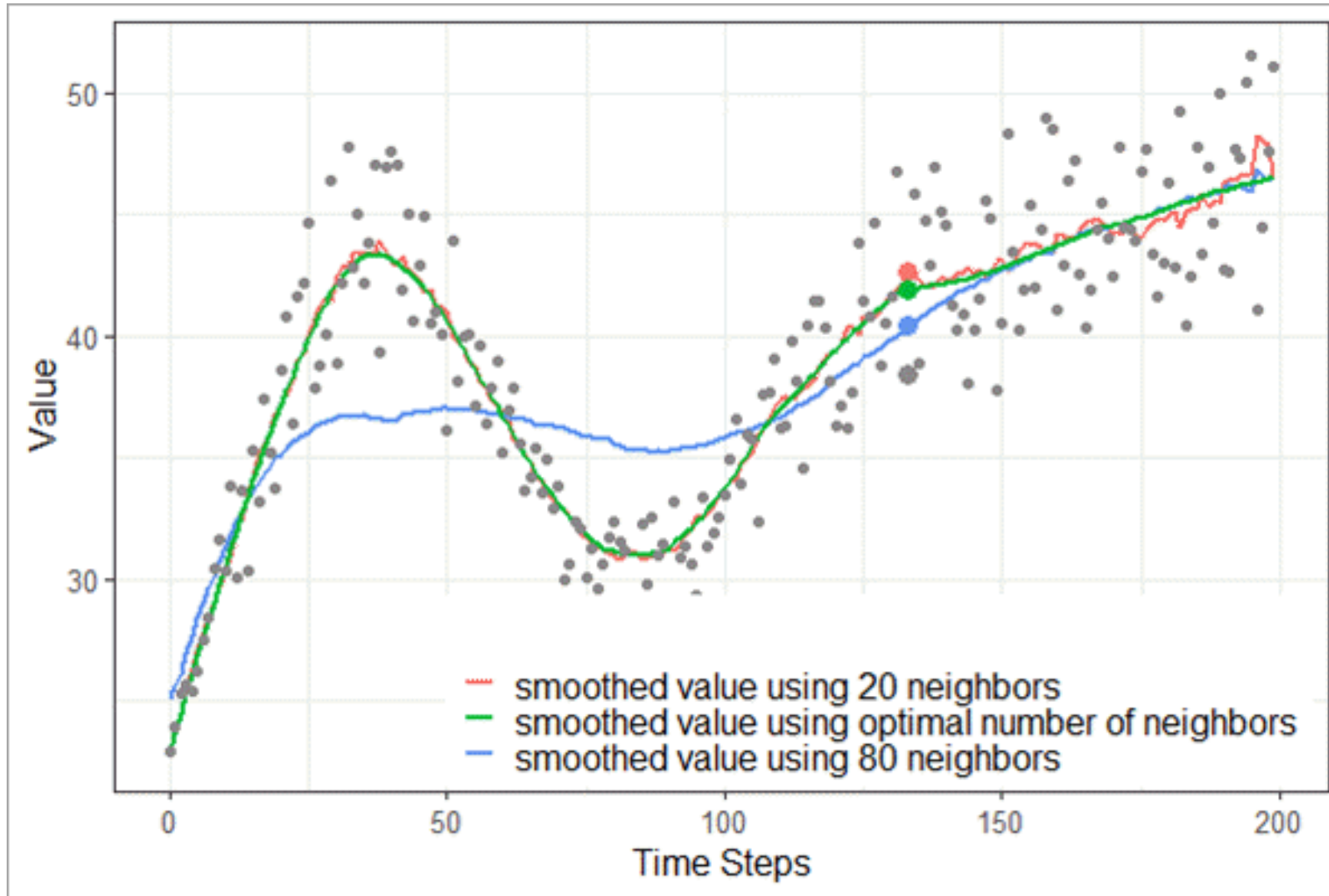
Time Series/Aggregate Analysis

- Monitoring longitudinal behavior, not just point-in-time metrics
 - Time-series approaches detect patterns, anomalies, and accumulative risks that indicate systemic vulnerabilities
 - Why time-series analysis matters
 - *Identifies slow-moving, chronic failures (e.g., growing patch backlog).*
 - *Reveals seasonality or cyclical patterns (e.g., transaction spikes causing stress).*
 - *Highlights drift in performance or control effectiveness.*
 - *Provides early warnings when multiple KRIs deteriorate simultaneously.*
 - Mature organizations aggregate metrics across:
 - *Business units (e.g., rising access exceptions across departments)*
 - *Systems (e.g., API latency trends across microservices)*
 - *Geographies (e.g., region-wide authentication failures)*
 - *Control families (e.g., decline in detective control success rates)*

Advanced Analytical Techniques

- Moving averages /exponential smoothing
 - Purpose
 - *Smooth noisy data to expose underlying trends and direction.*
 - Example in risk monitoring
 - *KRI: Failed login attempts per hour (indicator of brute-force attack or credential abuse)*
 - *Raw failed login counts are highly volatile*
 - *Using a 7-day moving average, a rising trend becomes visible even if daily values fluctuate*
 - *Exponential smoothing assigns higher weight to recent values, making emerging attacks easier to spot*
 - Outcome
 - *The SOC detects an upward directional trend in authentication anomalies*
 - *This triggers proactive controls like throttling, geo-blocking, or requiring MFA challenges*

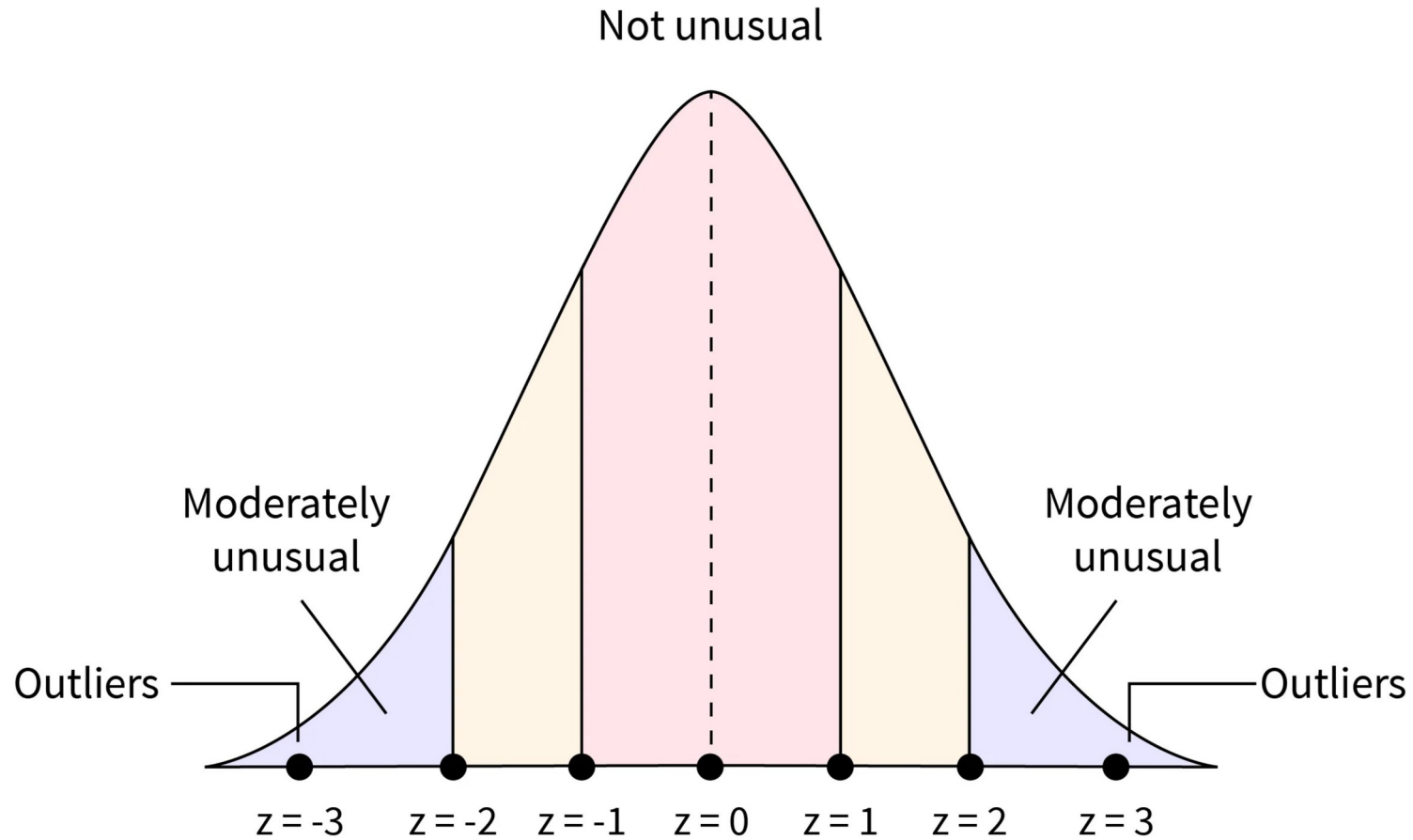
Smoothing



Advanced Analytical Techniques

- Z-Score Normalization / outlier detection
 - Purpose
 - *Identify values that deviate significantly from normal behavior*
 - Example in risk monitoring
 - *KRI: Outbound data transfer volume from a sensitive database*
 - *The system computes a Z-score for each hour's outbound traffic*
 - *Z-score > 3 (three standard deviations above the mean) triggers a security alert for potential data exfiltration*
 - *Even if the transfer size is below an absolute threshold, statistical deviation flags the anomaly*
 - Outcome
 - *An insider threat or compromised account can be detected before significant data is lost, based solely on abnormal behavior relative to historical patterns*

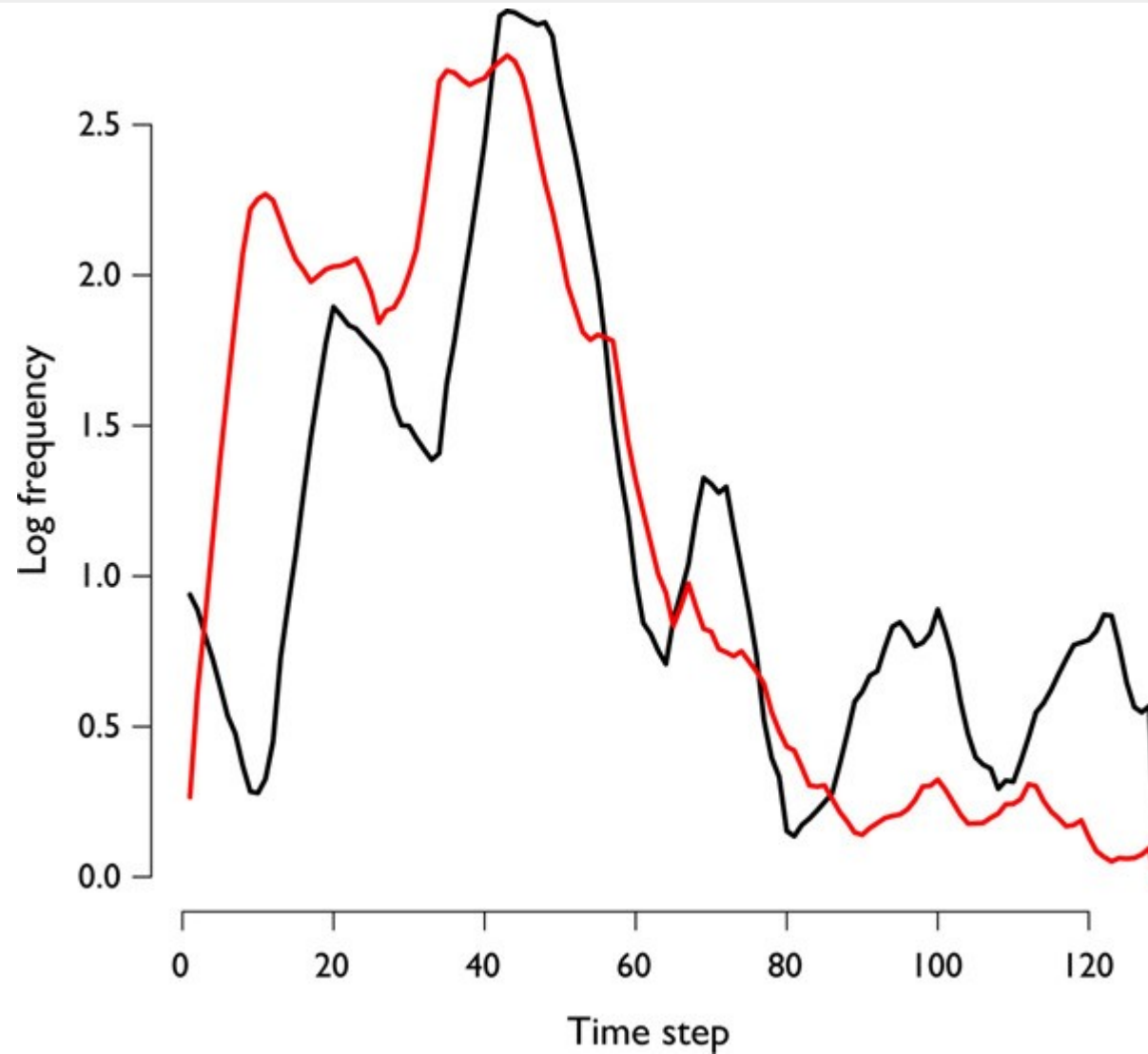
Detecting Outliers with z-Scores



Advanced Analytical Techniques

- Lag analysis
 - Purpose
 - *Measure whether changes in one metric precede changes in another metric*
 - Example in risk monitoring
 - *KPI: Patch completion rate vs KRI: Count of critical vulnerabilities*
 - *Lag-7 analysis reveals*
 - *A decrease in patch completion today correlates with*
 - *An increase in critical vulnerabilities seven days later*
 - Outcome
 - *Risk teams can demonstrate a causal or predictive relationship between a KIR and heightened vulnerability exposure which strengthens the case for resource allocation or process redesign*

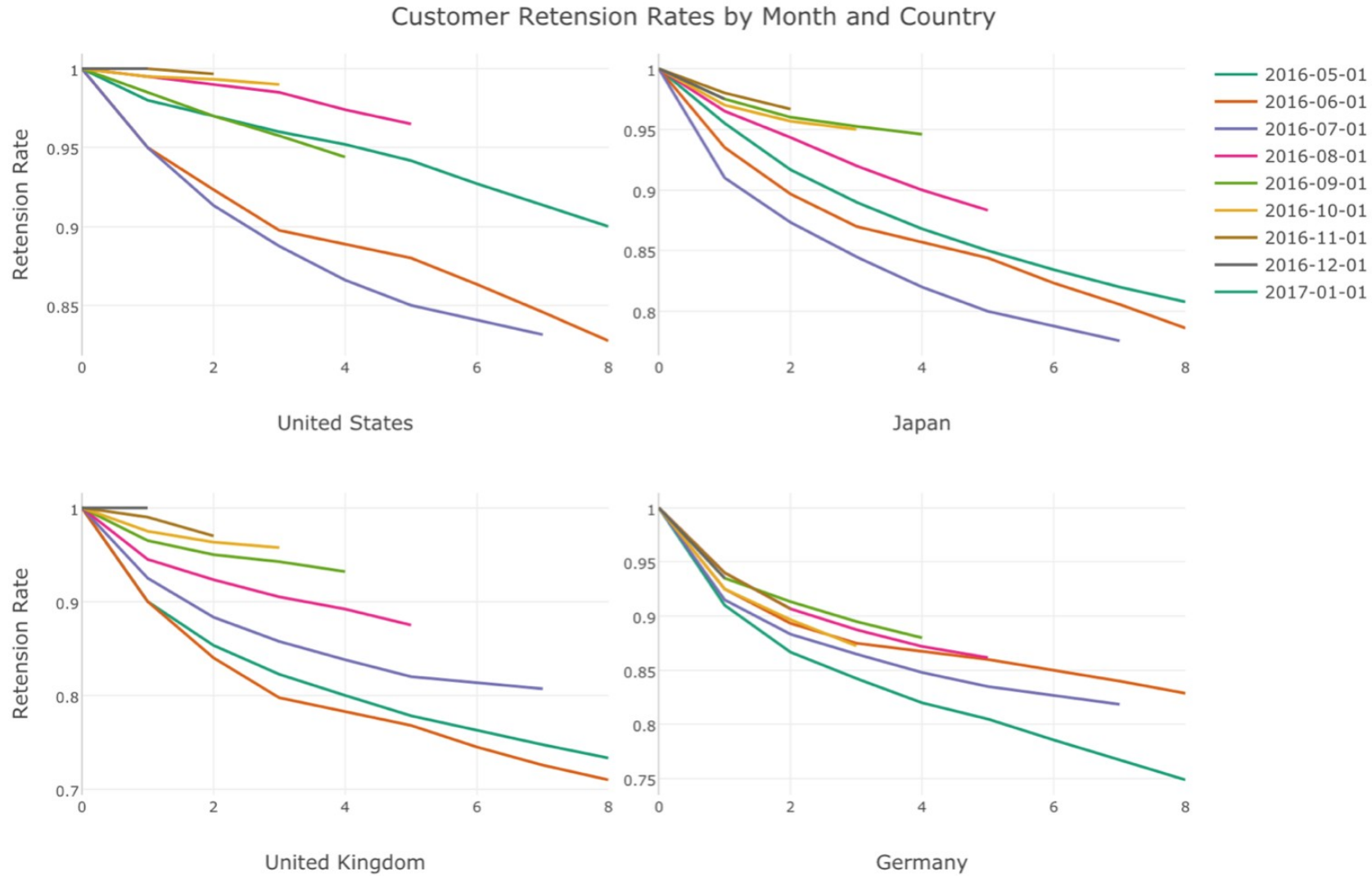
Lag Analysis



Advanced Analytical Techniques

- Cohort trend analysis
 - Purpose
 - *Compare similar groups (cohorts) to find patterns or outliers within a category*
 - Example in risk monitoring, cohorts are created by
 - *Application type (web apps vs. batch apps)*
 - *Technology stack (Java vs. Node vs. Python)*
 - *Environment (production vs. staging)*
 - *Ownership (Team A vs. Team B)*
 - A cohort analysis reveals
 - *Systems owned by Team B show a 3× higher growth rate of high-severity vulnerabilities compared to similar systems owned by other teams.*
 - Outcome
 - *This reveals a team-specific process breakdown rather than a global security weakness suggests targeted training, staffing, or process intervention can be deployed*

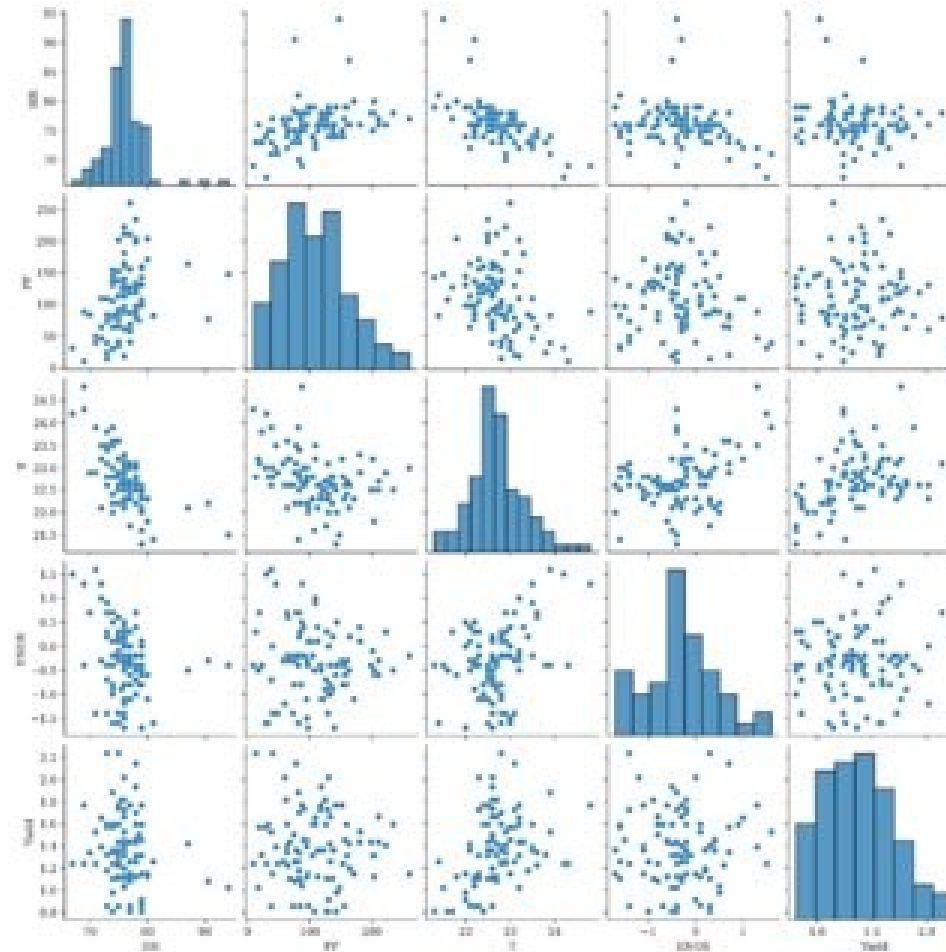
Cohort Analysis



Advanced Analytical Techniques

- Cross-correlation matrices
 - Purpose
 - *Identify how metrics influence each other across time.*
 - Metrics include
 - *KPI: Mean Time to Recover (MTTR) for incidents*
 - *KRI: Incident frequency*
 - *KPI: Change success rate*
 - *KRI: Number of repeated near misses*
 - Cross-correlation shows
 - *When change success rate decreases*
 - *Incident frequency increases with a correlation coefficient of 0.82*
 - *Additionally, higher MTTR correlates with a rise in repeated near misses, suggesting control inefficiency*
 - Outcome
 - *The correlation matrix grounds the argument that unstable change processes are the root cause of operational risk increases which should trigger major change-control reforms*

Cross-Correlation Matrices

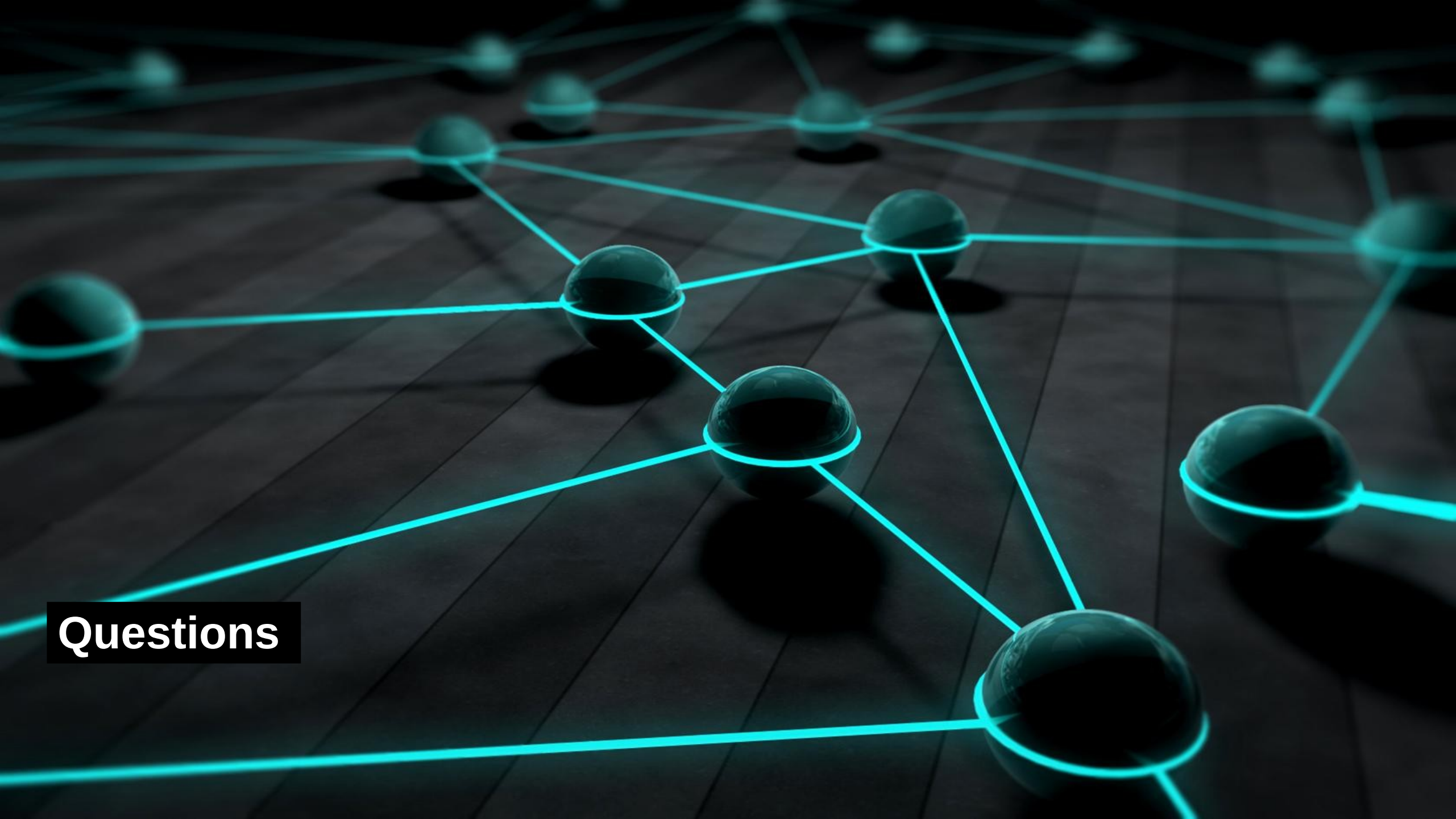


Advanced Analytical Techniques

- Composite KRI Index
 - Purpose
 - *Combine disparate KRIs into a single normalized risk score that can be tracked over time*
 - Combine KRIs such as
 - *Number of critical vulnerabilities (weighted 30%)*
 - *Privileged access exceptions (20%)*
 - *Failed login anomalies (20%)*
 - *Backup integrity failures (15%)*
 - *End-of-life infrastructure ratio (15%)*
 - Outcome
 - *The correlation matrix grounds the argument that unstable change processes are the root cause of operational risk increases which should trigger major change-control reforms*

Advanced Analytical Techniques

- Composite KRI Index
 - Using normalized values and weights produces a Composite KRI Index (0–100).
 - Example trend
 - Q1: 42
 - Q2: 49
 - Q3: 63
 - Q4: 71
 - Even without a major incident, the index shows a clear erosion of risk posture, prompting leadership to invest in remediation before conditions worsen
 - Composite indices provide:
 - *Board-friendly reporting*
 - *Cohesive trend visibility*
 - *A single metric that captures complex, multidimensional risk exposure*
 - *This enables strategic prioritization and budget planning*



Questions